



SKG14 IC8 Pacing and Emerging Technologies for ICM Engineers, L2

3.0 Key Technologies of IIoT

July 2025

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Records of Change

Date of Issue/ Reissue	Revision No.	Reason of Development / Change
July 2025	00	New module offering inline with TI&R 2025.

Course Content

3.1	Next Generation Sensors	7
	Smart Sensors Architecture	
	Smart Sensors Characteristics	
3.2	IIoT Gateways	20
	IIoT Gateways vs. Existing OT Gateways	
	Key Advantages	
3.3	IIoT Edge Compute Devices	27
	OT Edge Devices	
	IIoT Edge Compute Devices	
3.4	Next Gen Controllers (IIoT Enable)	31
	OT Controllers	
	IIoT Controllers	
3.5	Cloud Services	38
	Current OT Implementation	
	Common IIoT Cloud Services	
3.6	IIoT Connectivity (IIoT Protocol)	44
	Common OT Protocols	
	IIoT Protocols	
3.7	IIoT Integration Middleware	52

Learning Outcome

At the end of the session, you will be able to:

- Understand the advancements in sensor technology and their impact on industrial applications.
- Explain the role of IIoT gateways in enabling seamless communication between devices and networks.
- Describe the importance of edge computing in IIoT and how it enhances real-time data processing.
- Explore the evolution of industrial controllers and their integration with IIoT systems.
- Understand the significance of cloud computing in IIoT, including data storage, analytics, and scalability.
- Explain the communication protocols that facilitate secure and efficient IIoT connectivity.
- Discover how middleware solutions streamline IIoT integration across diverse systems and applications.

03. Key Technologies of IIoT

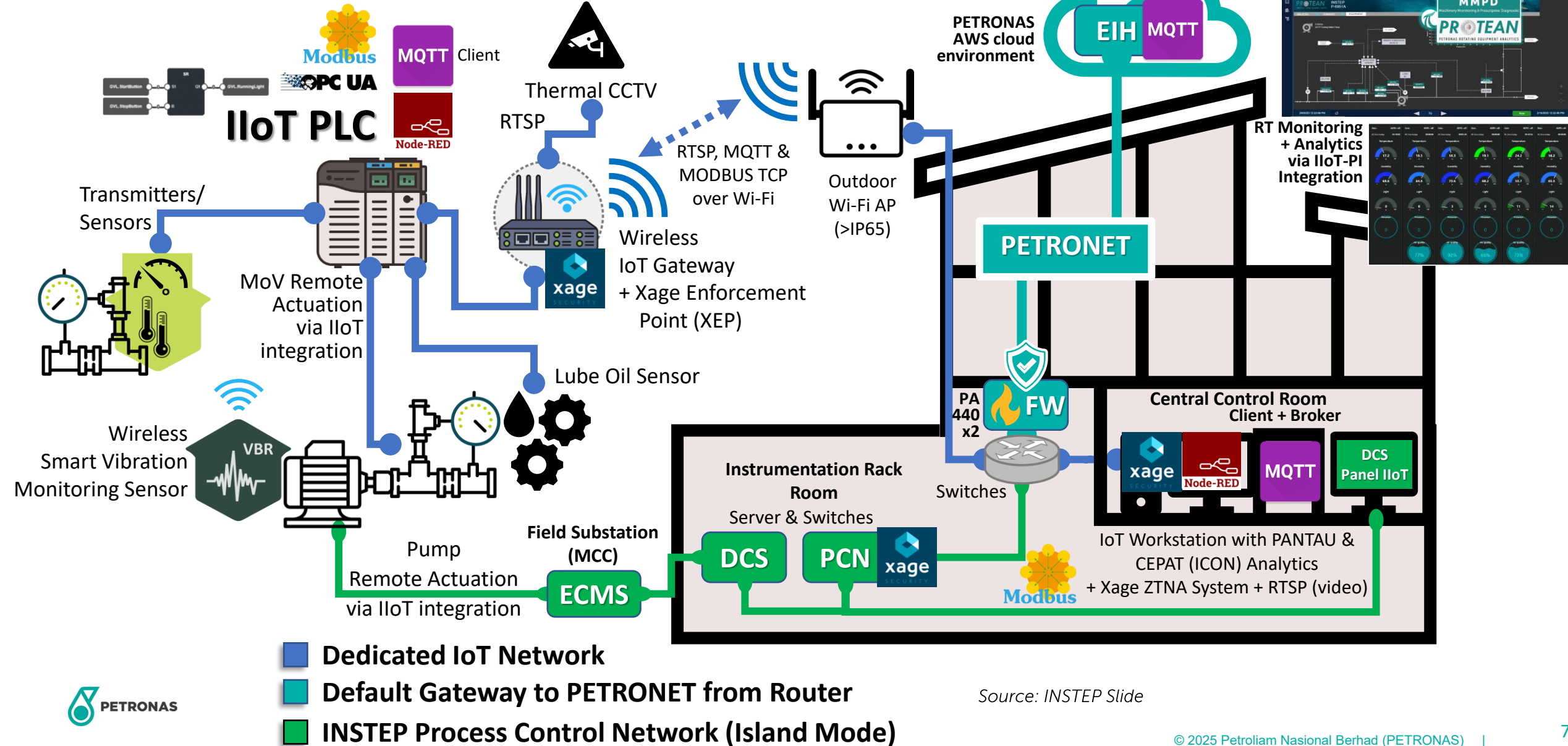
- Next Generation Sensors
- IIoT Gateways
- IIoT Edge Compute Devices
- Next Gen Controllers (IIoT Enable)
- Cloud Services
- IIoT Connectivity (IIoT Protocol)
- IIoT Integration Middleware

03. Key Technologies of IIoT

- Next Generation Sensors
- IIoT Gateways
- IIoT Edge Compute Devices
- Next Gen Controllers (IIoT Enable)
- Cloud Services
- IIoT Connectivity (IIoT Protocol)
- IIoT Integration Middleware

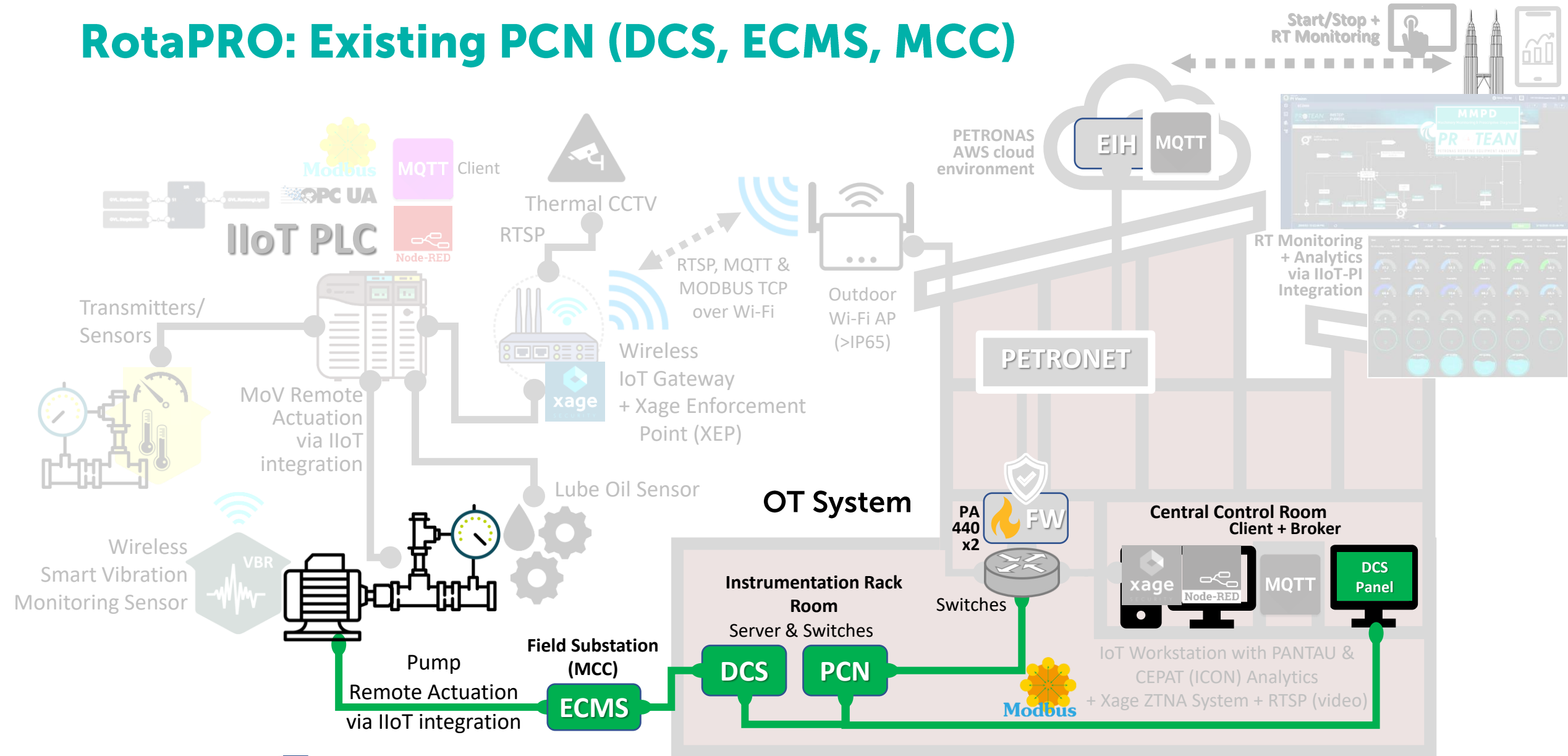
RotaPRO (Rotating Package Remote Operation)

INSTEP for Remote Monitoring and Control



Source: INSTEP Slide

RotaPRO: Existing PCN (DCS, ECMS, MCC)

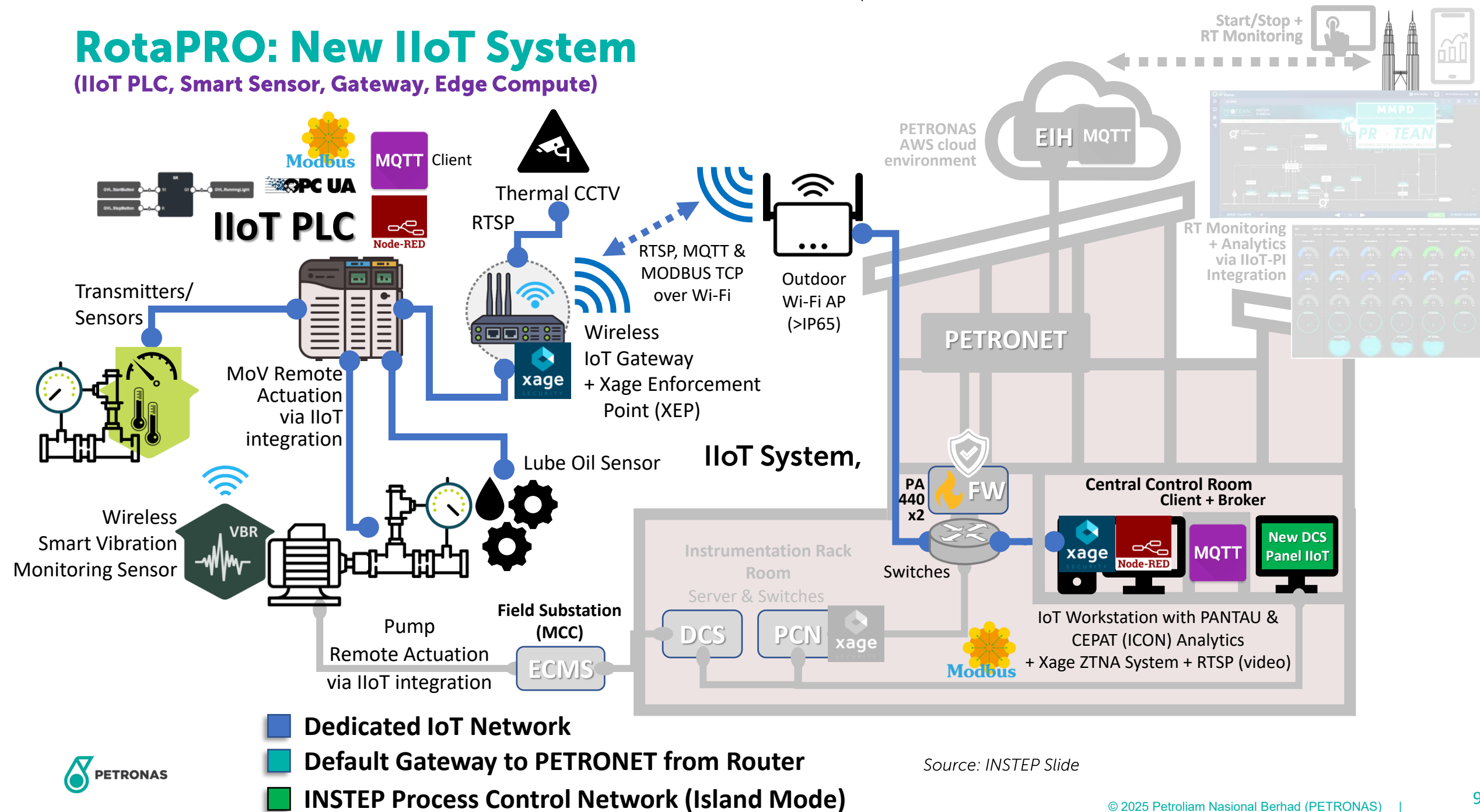


- **Dedicated IoT Network**
- **Default Gateway to PETRONET from Router**
- **INSTEP Process Control Network (Island Mode)**

Source: INSTEP Slide

RotaPRO: New IIoT System

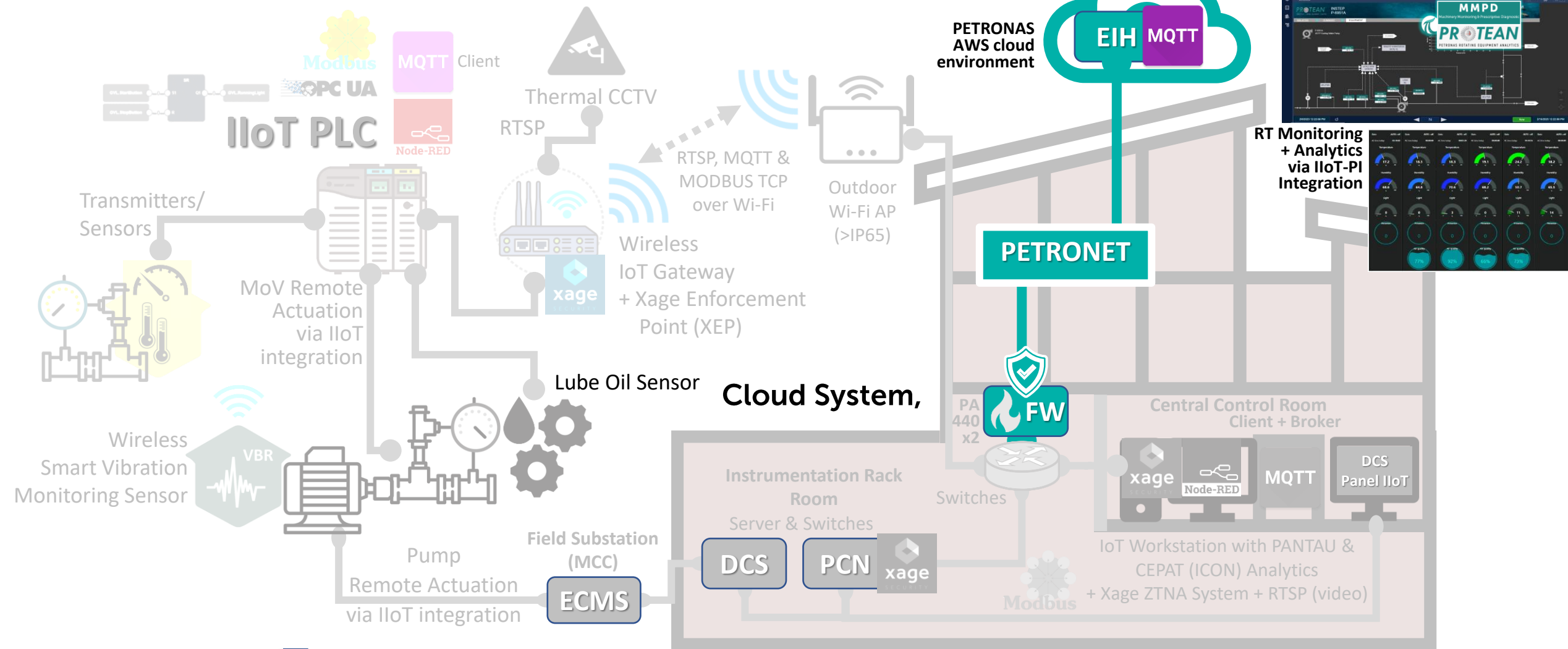
(IIoT PLC, Smart Sensor, Gateway, Edge Compute)



Source: INSTEP Slide

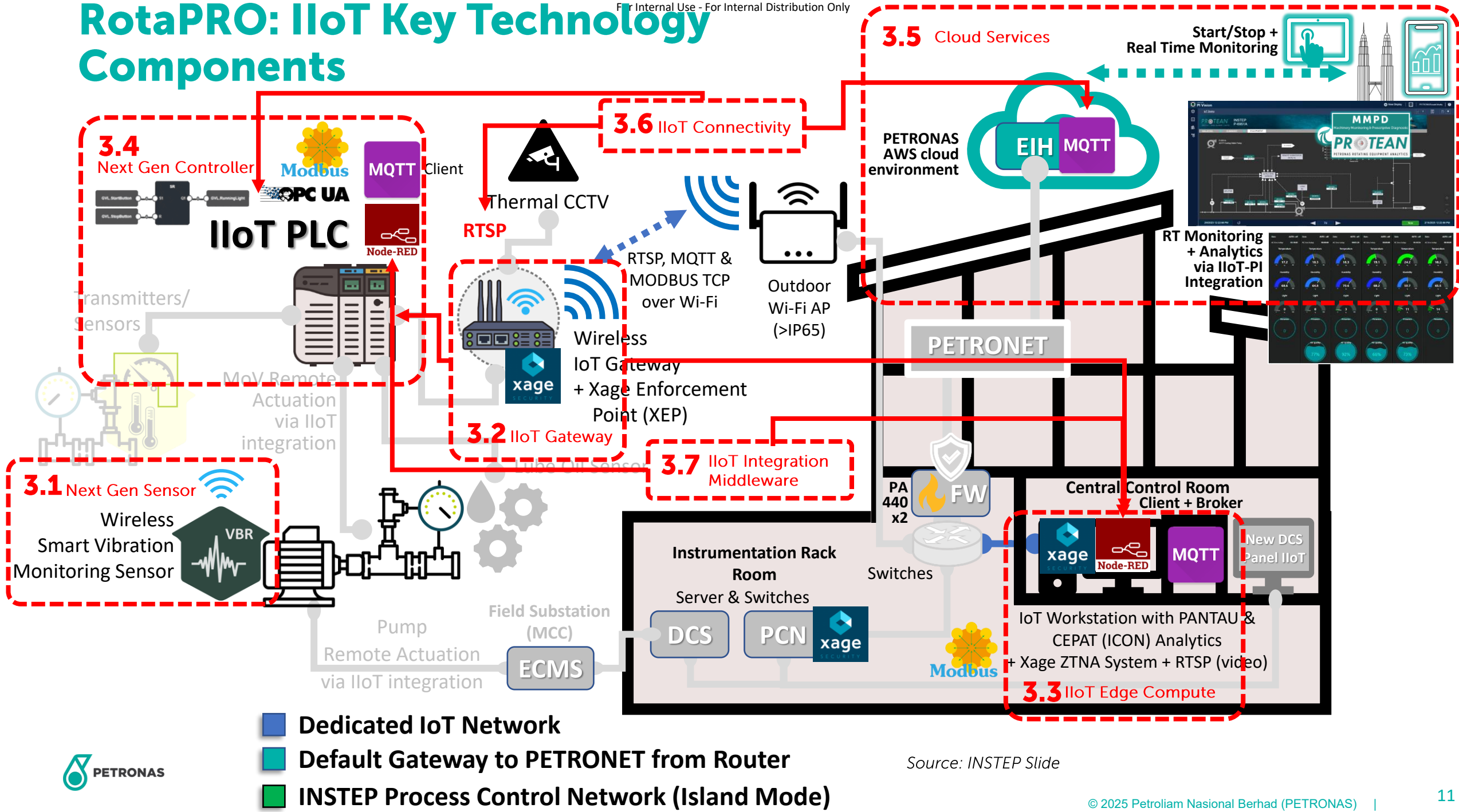
RotaPRO: Cloud System

(AWS MQTT Broker, write to PI Tag, PI system, PROTEAN, MMPD)



- **Dedicated IoT Network**
- **Default Gateway to PETRONET from Router**
- **INSTEP Process Control Network (Island Mode)**

RotaPRO: IIoT Key Technology Components



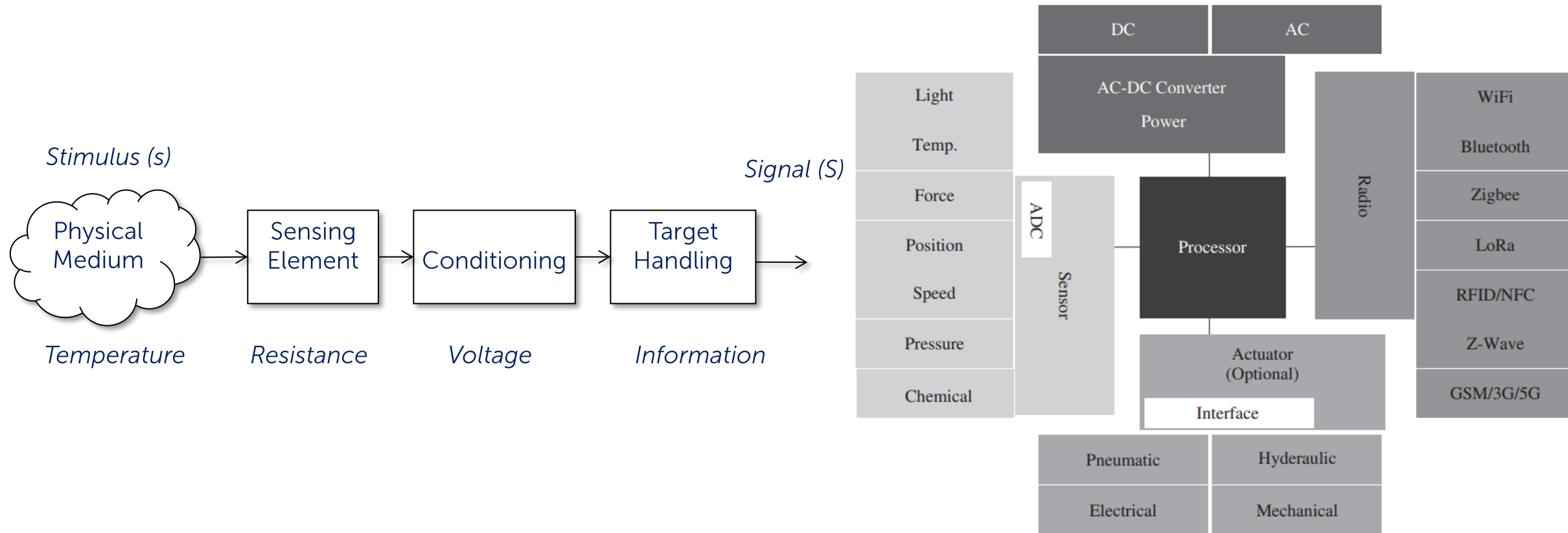
Next Generation Sensors

Smart Sensors



Next Generation Sensors

Smart Sensors Architecture



Next Generation Sensors

Next-generation sensors represent a leap forward in industrial automation and monitoring by integrating advanced technologies such as edge computing, artificial intelligence (AI), and wireless communication.

These sensors go beyond simple data collection, offering real-time processing and analysis at the source.

They enable predictive maintenance, fault detection, and operational optimization by analyzing data patterns and triggering actions without needing centralized processing.

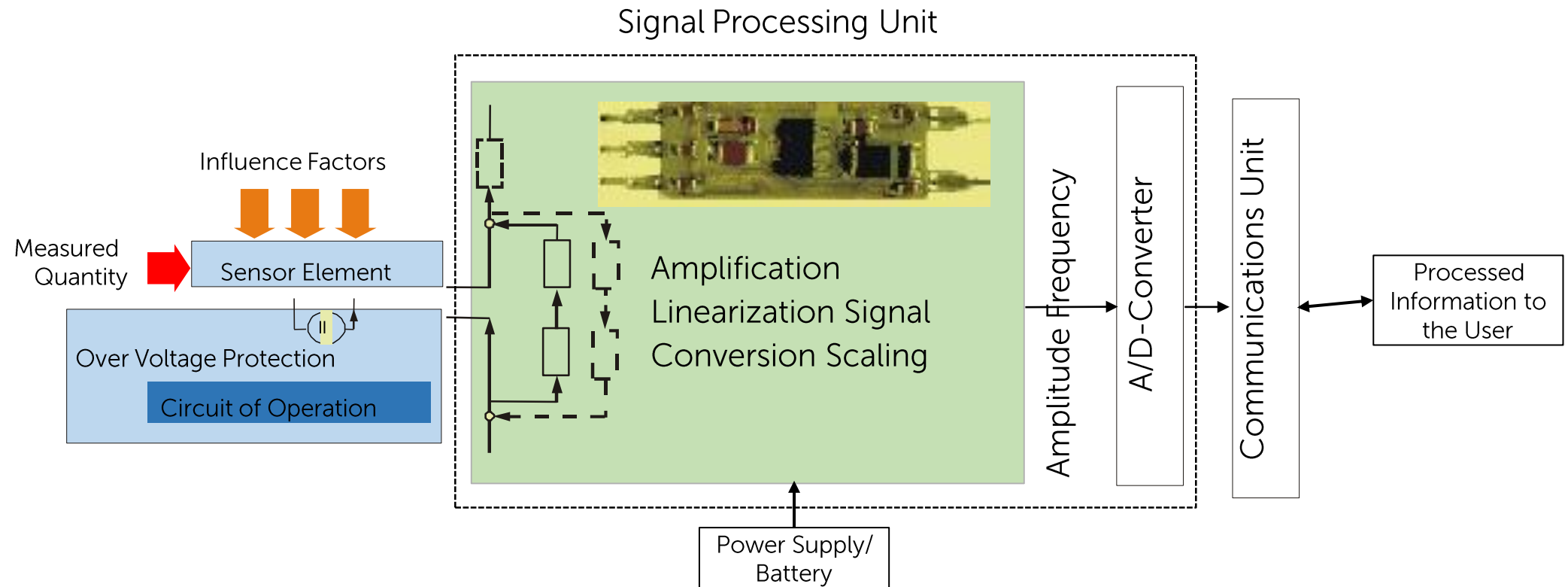
A key characteristic of next-generation sensors is their ability to support wireless communication protocols such as 5G, LoRaWAN, and Bluetooth Low Energy (BLE).

Additionally, modern sensors feature power-efficient designs and can use energy harvesting techniques, ensuring long operational lifespans even in challenging conditions.

Cybersecurity is another crucial aspect of next-generation sensors, which incorporate secure communication protocols, encrypted data transmission, and tamper-resistant hardware to safeguard sensitive industrial data.

Next Generation Sensors

Smart Sensors Architecture



Next Generation Sensors

Capabilities

Smart Sensing Capabilities

Wireless Connectivity

Energy Efficiency

Self-diagnostics and
Predictive Maintenance

AI and Machine Learning
Integration

Edge Computing Support

Next Generation Sensors

Smart Sensors Characteristics

Static characteristics

- Accuracy
- Resolution
- Precision
- Expected value
- Error
- Sensitivity

Dynamic characteristics

- Zero order systems
- First order systems
- Second order systems

Next Generation Sensors

Traditional Sensors:

- Standalone, limited functionality, analog signals.

IIoT Sensors:

- Real-time, high-accuracy, advanced features like anomaly detection and computer vision.

Key Advantage:

- Enable remote diagnostics, real-time decision-making, and advanced monitoring capabilities.

Examples:

- Ranger Pro, Yokogawa Sushi Sensor, Pepperl+Fuchs IO-Link System, RFID/RTLS (Asset & People tracking), etc.

03. Key Technologies of IIoT

- Next Generation Sensors
- **IIoT Gateways**
- IIoT Edge Compute Devices
- Next Gen Controllers (IIoT Enable)
- Cloud Services
- IIoT Connectivity (IIoT Protocol)
- IIoT Integration Middleware

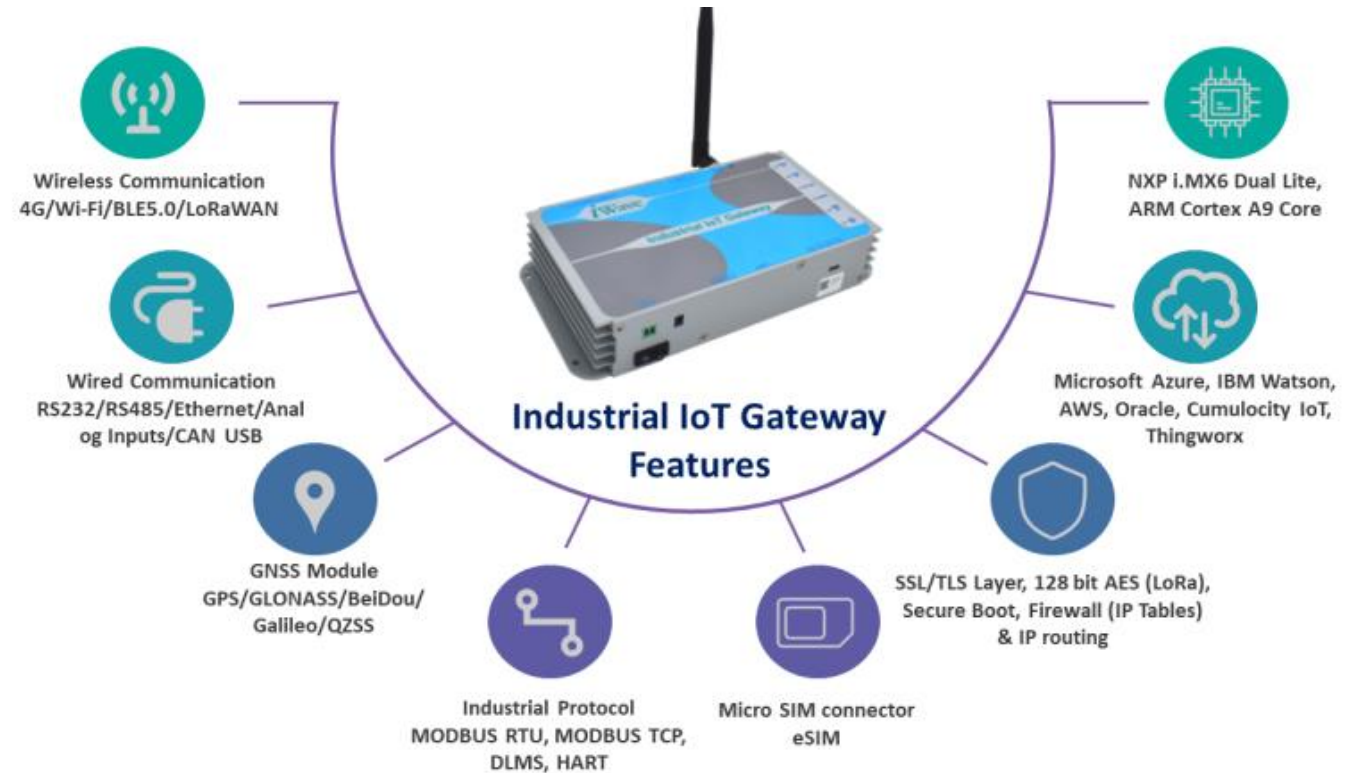
IIoT Gateways

- Operational Technology (OT) Gateways:

Dedicated protocol conversion between legacy OT protocols (e.g., Modbus, Hart, Profibus, etc), and enabling integration to IT networks

- IIoT Gateway:

Provide secure multi-network connectivity and advanced data handling, real-time data processing with analytics, multi-protocol integration (OT-IIoT-IT) and cloud services integration.



Sources: <https://www.iwavesystems.com/news/industrial-iot-gateways-for-enterprise-industry-4-0-and-energy-monitoring/>

IloT Gateways

IloT Gateways vs. Existing OT Gateways

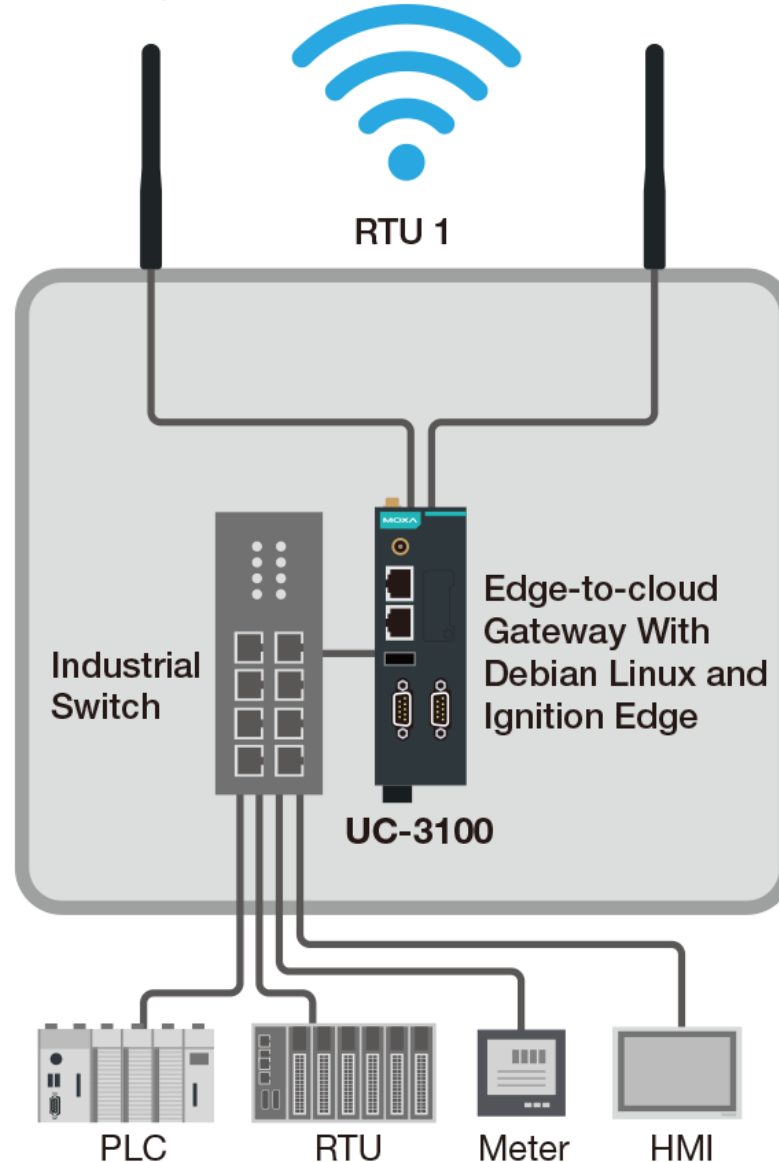
OT Gateways	Key Differences	IloT Gateways
OT Gateways focus on real-time communication and protocol conversion within the OT environment, enabling data transfer between legacy OT systems and IT networks.		IloT Gateways also perform real-time data processing, but they go further by enabling advanced analytics at the edge, integrating multiple systems (OT, IT, IloT), and providing cloud connectivity for more complex decision-making and optimization.
OT Gateways: • Function: Dedicated protocol conversion between legacy OT protocols (e.g., Modbus, Profibus, HART) and integration to IT systems (e.g., OPC, PI). • Example OT Gateways: Emerson DeltaV™ Communicator, Rockwell Automation Allen-Bradley Stratix™ 5000 and 5700 Series, Moxa UC-8100 Series, Schneider Electric Modicon M580, Siemens SCALANCE X-200 and X-300, Advantech WISE-4000 and WISE-710, • Key Features: ◦ Real-time data processing (e.g., filtering, aggregation). ◦ Protocol conversion (e.g., Modbus to Ethernet, Profibus to TCP/IP). ◦ Limited scalability, focusing on basic communication and data forwarding.		IloT Gateways: • Function: Enable secure multi-network connectivity, advanced data handling, real-time data processing with analytics, and cloud integration. • Example IloT Gateway: Siemens IOT2040, Moxa UC-8100, Advantech WISE-4000, Emerson ROC, Phoenix Contact FL Gateway, Red Lion Data Station Plus. • Key Features: ◦ Real-time data processing with advanced edge analytics. ◦ Multi-protocol integration (e.g., Modbus, OPC-UA, MQTT). ◦ Cloud services integration, edge analytics, and multi-network communication.

IIoT Gateways

Key Advantages

- **Advanced Analytics at the Edge:** IIoT Gateways can perform edge computing and real-time analytics, enabling smarter decision-making without needing to send data to central systems.
- **Cloud Integration:** IIoT Gateways support seamless connection to cloud platforms for data storage, remote monitoring, and AI/ML integration.
- **Multi-Protocol Integration:** IIoT Gateways support multiple protocols (e.g., Modbus, MQTT, OPC-UA), facilitating easier communication across OT, IT, and IIoT systems.
- **Scalability and Flexibility:** IIoT Gateways enable more scalable solutions by connecting varied devices, enabling cross-system integration, and remote configuration.
- **Edge Decision-Making:** With local data processing, IIoT Gateways can make real-time decisions that enhance automation and efficiency at the edge without latency from central systems.

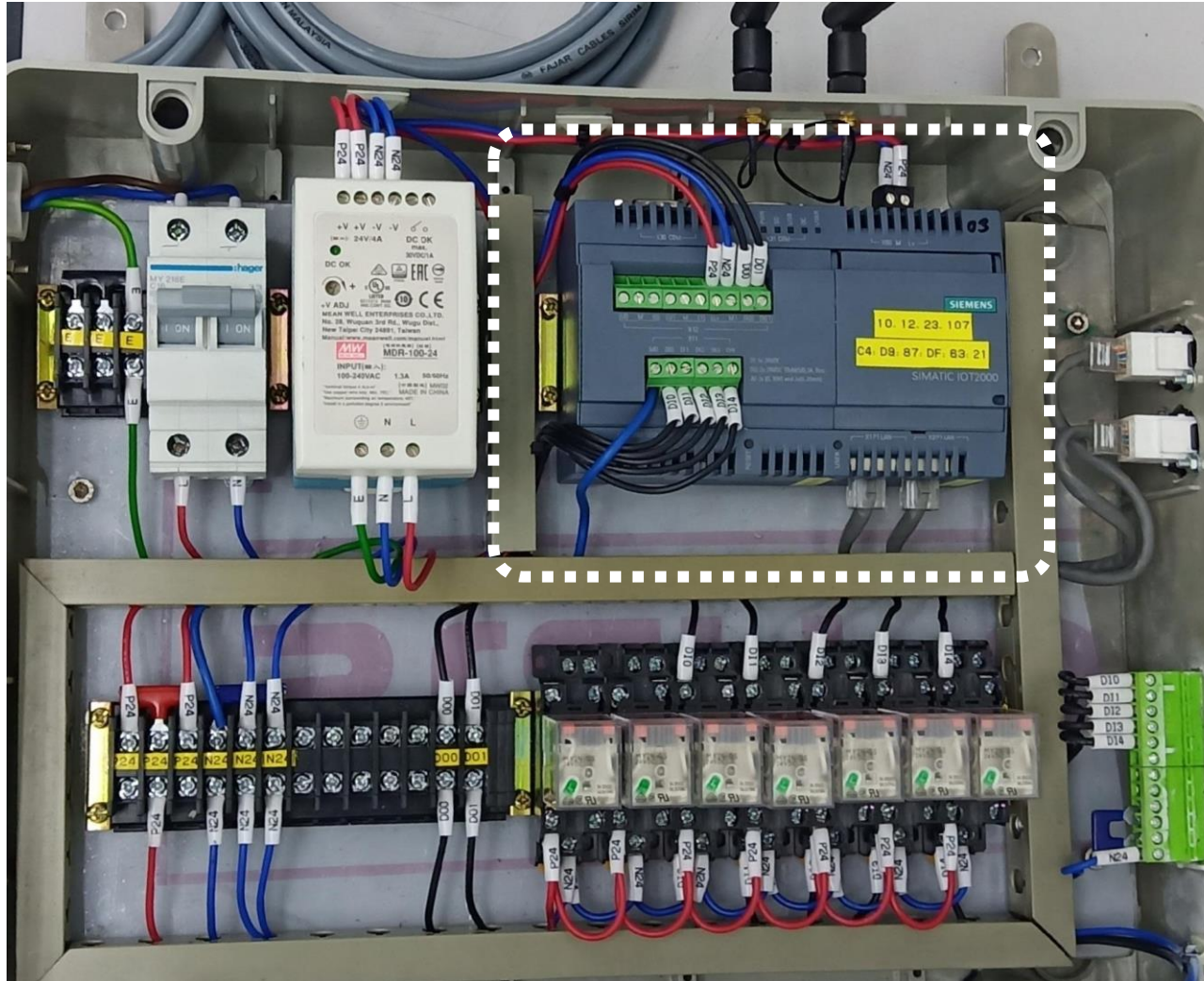
IIoT Gateways – Local Panel Example



- The IIoT Gateway is connected to various types of devices (Ex. PLC, RTU, Meter, HMI, etc.)
- Each device may be using different protocols (Ex. MODBUS, OPC classic, OPC-UA, WIRELESS HART, etc.)
- All devices has unique IP address and are reachable via Ethernet or Wi-Fi network
- IIoT Gateway has an Operating System that can run related applications, like Ignition, Kepware, Node Red, etc. to enable multi-protocol data integration, aggregation
- Send processed data using MQTT protocol to a remote central server, OR on-prem edge computer before sending to cloud, OR direct to cloud hosted IIoT system, for monitoring, analytics, reporting, and alert/notifications

IIoT Gateways

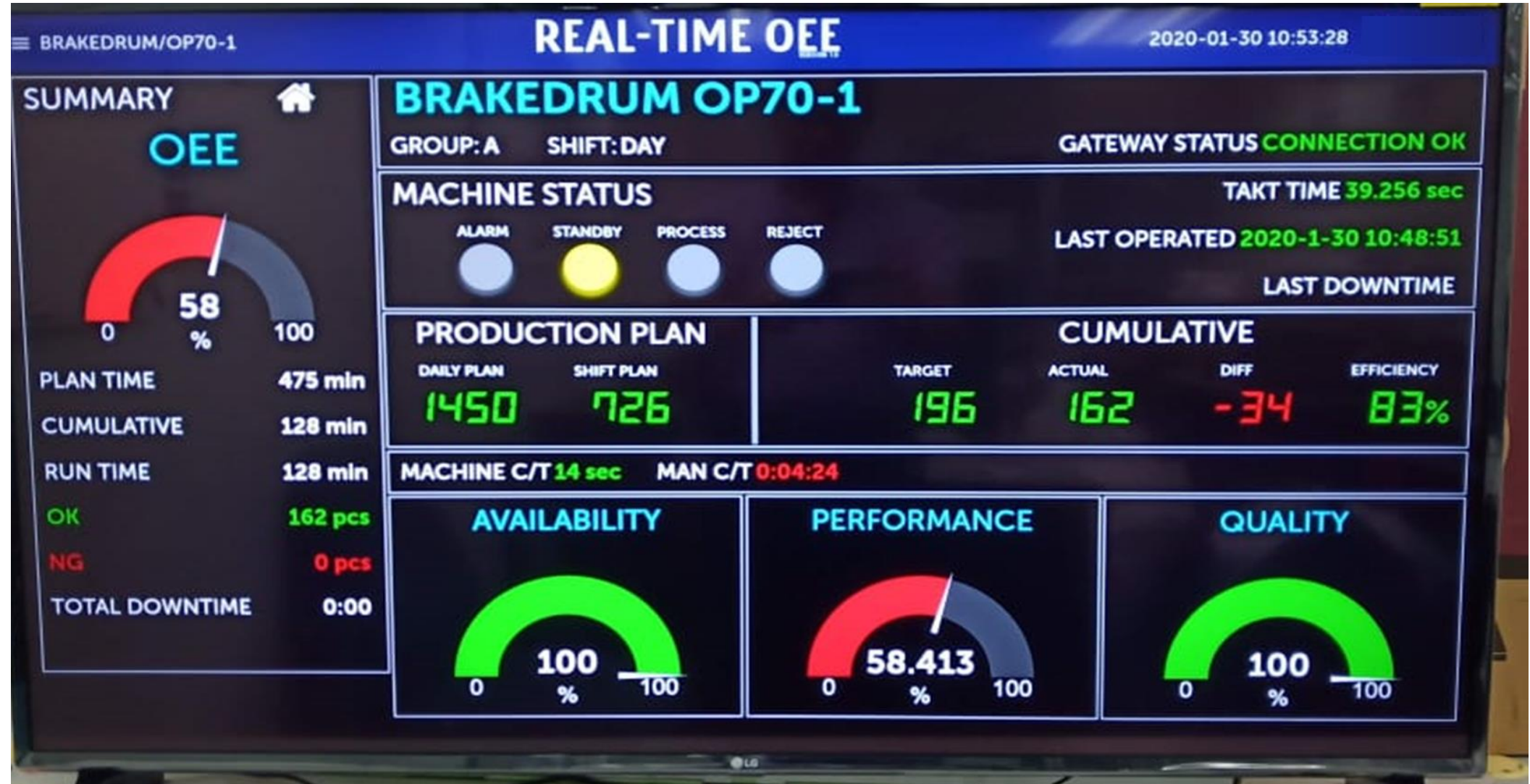
OEE Example



- Here is an example of an IIoT Gateway with an installed Digital Input/Output module using 0-24V signals for data processing to enable Overall Equipment Efficiency (OEE) in real-time
- Directly processes the Digital Inputs from sensors placed in the production line to enable:-
 - Real-time data processing within the IIoT Gateway
 - using MQTT over Wi-Fi network to forward data to a central system for
 - data aggregation
 - Analysis
 - OEE Dashboarding
 - alert/ notifications to respective production/technical team

IIoT Gateways

Real-Time OEE Dashboard Example



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- **IIoT Edge Compute Devices**
- Next Gen Controllers (IIoT Enable)
- Cloud Services
- IIoT Connectivity (IIoT Protocol)
- IIoT Integration Middleware

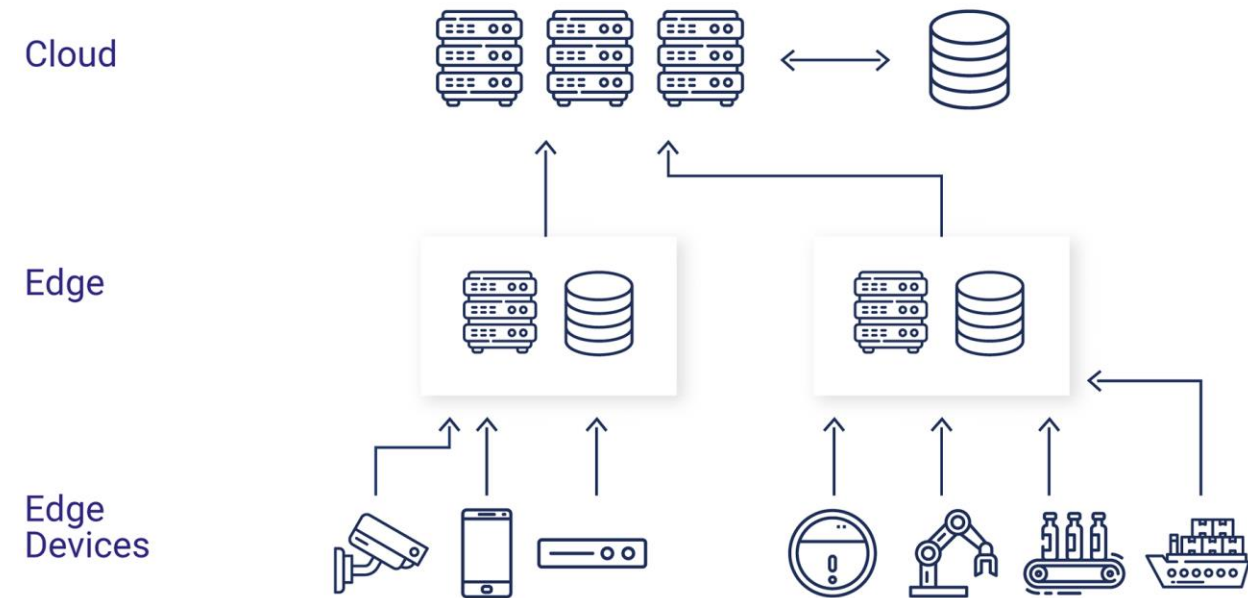
IIoT Edge Compute Devices

- OT Edge Device: (Ex. DCS, PLC, RTU, etc.)

Focus on real-time control, data acquisition, and automation using legacy OT protocols (Ex. Modbus, OPC, Profibus). Primarily for local control and communication within industrial environments.

- IIoT Edge Compute Devices:

Provide local data processing to minimize latency, real-time decision-making, and advanced analytics at the edge. Support multi-network integration and cloud connectivity for smarter, data-driven insights.



Source: <https://www.3pillarglobal.com/insights/blog/edge-computing-and-iiot/>

IIoT Edge Compute Devices

- **IIoT Edge Compute Devices:**

Purpose-built for processing data at the edge of industrial networks. These devices enable real-time analytics, protocol translation, lightweight AI/ML, and secure communication with IT/cloud systems, often operating alongside or in support of DCSs/PLCs/RTUs.

- **Role in IIoT Architectures:**

Act as the bridge between physical assets (via OT protocols) and enterprise systems. They can ingest, clean, transform, and analyze data without relying on centralized cloud systems, improving latency, bandwidth use, and decision-making speed.

- **Key Advantages:**

- Local data processing and contextualization
- Reduced cloud dependency and bandwidth usage
- Supports containers, Node-RED, Python, MQTT, OPC UA, etc.
- Enables predictive analytics and ML inference at the edge
- Facilitates secure OT-to-IT integration
- Built to operate in rugged/remote environments

- **Examples:**

Stratus ztC Edge, HPE Edgeline, Siemens SIMATIC Industrial Edge Devices, Dell Edge Gateway, Opto22 groov EPIC, Sensia HCC2, OnLogic Rugged IPCs, Revolution Pi

IIoT Edge Compute Devices

Edge Compute Device or “Edge Computer” is a functional term, not just a hardware form factor.

A device qualifies as an **edge computer** if it's deployed close to the data source (on-site, near machines, processes, or sensors) and **performs compute tasks locally**, such as (but not limited to) :-

- Data preprocessing
- Analytics
- Protocol conversion
- Visualisation
- Computer Vision/ AI/ ML inference
- Real-time decision-making

Edge Compute Form Factor	Typical Use Case	Compute Capability
Mini PCs / NUCs	Protocol conversion (e.g., Modbus → MQTT), lightweight dashboards, device health logic	Low
IIoT Gateways	Data preprocessing, Node-RED flows, MQTT brokers, local logic orchestration	Low–Medium
Workstations	Local SCADA, HMI panels, mid-level analytics, simple ML (e.g., anomaly detection)	Medium
Ruggedized IPCs	Real-time historian, vibration analytics, OPC UA servers, AI/ML inference models	Medium–High
Rackmount Servers / Industrial Edge Servers	On-prem AI pipelines, computer vision, GenAI models, AI chatbots, semantic search, private LLMs	High–Very High
Edge Compute Form Factor	Typical Use Case	Compute Capability

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- **Next Gen Controllers (IIoT Enable)**
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- IIoT Integration Middleware

Next Generation Controllers (IIoT Enabled)

- OT Controllers:

Focused on real-time control, automation, and data acquisition within industrial systems, typically using legacy OT protocols (e.g., Modbus, Profibus), designed for local control and monitoring

- IIoT Enabled Controllers:

Build on traditional OT controllers with enhanced connectivity, real-time decision-making, and advanced analytics. They serve as intelligent controllers at the edge, facilitating OT-IT integration and supporting cloud services for data-driven insights

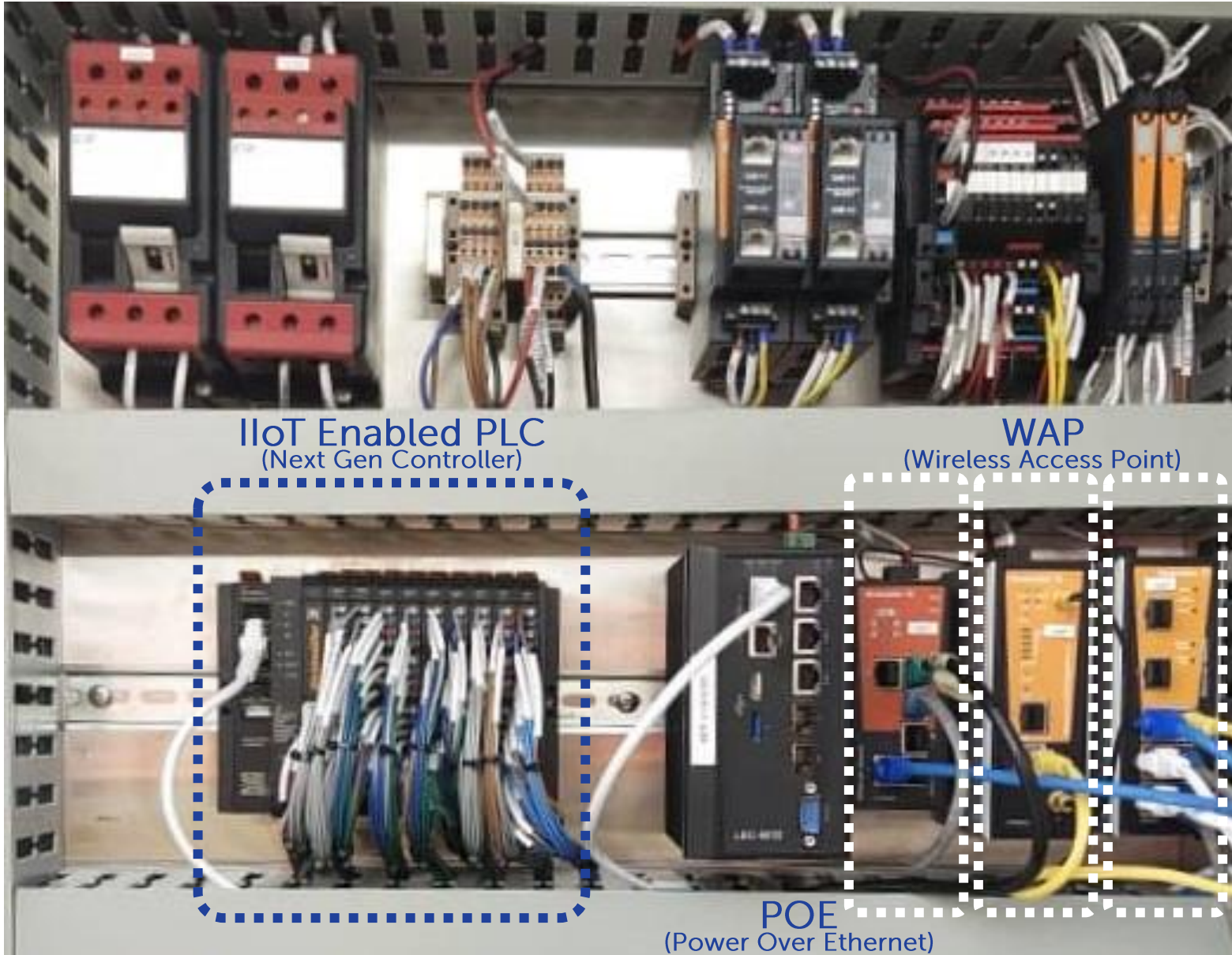
- Key Advantage:

Real-time decision-making, advanced edge analytics, multi-protocol integration (OT, IT, IIoT), cloud connectivity for remote monitoring, cross-system integration, and the ability to bridge OT and IT for smarter operations

- Examples:

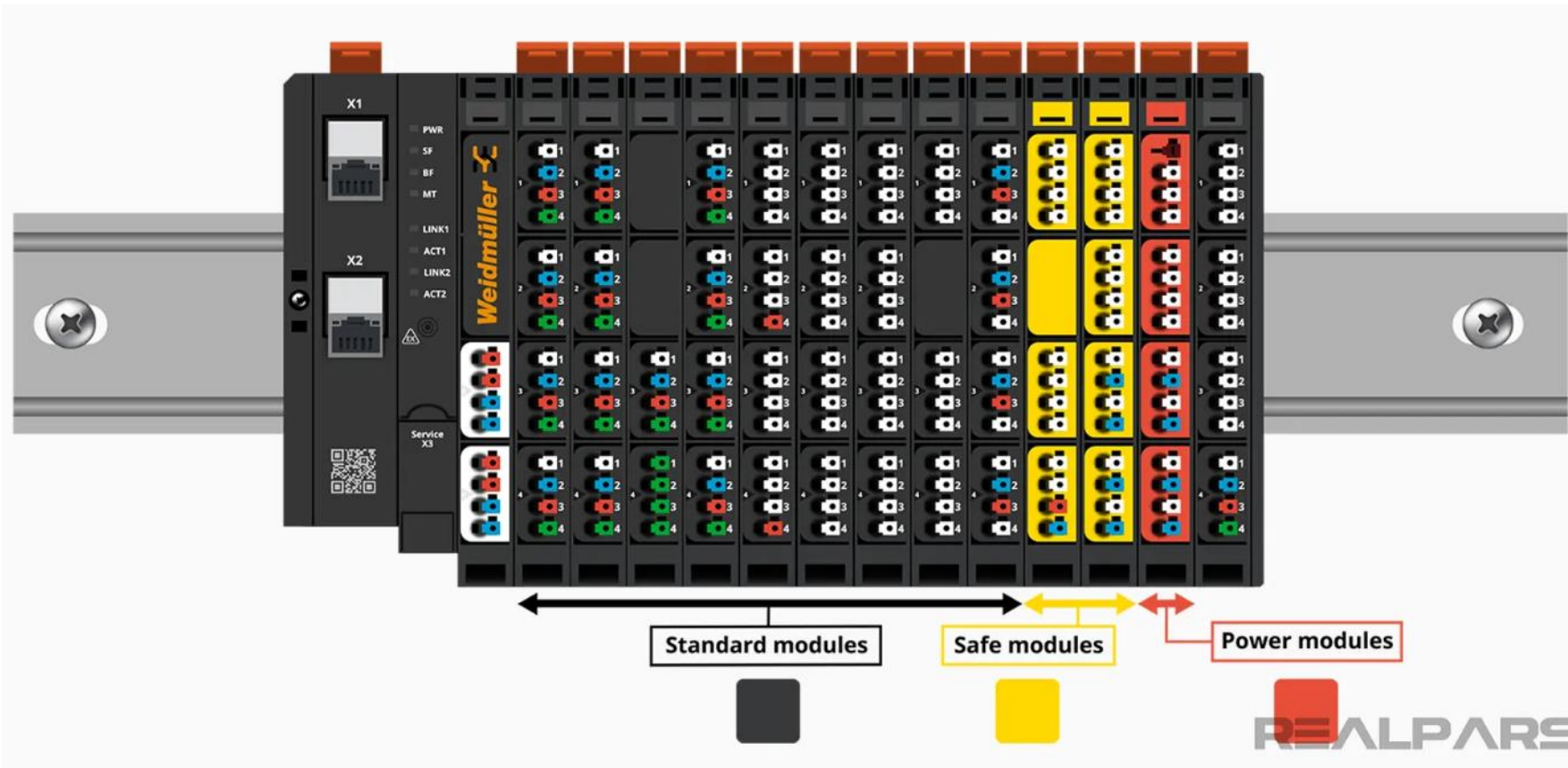
Weidmueller IIoT PLC, Siemens IOT2040/2050, Advantech IIoT Controllers, Industrial PC-based Controllers

Next Generation Controllers (IIoT Enabled)



Here is an example of an IIoT Enabled PLC deployed within a local panel, as part of the key enabler component for the RotaPRO solution.

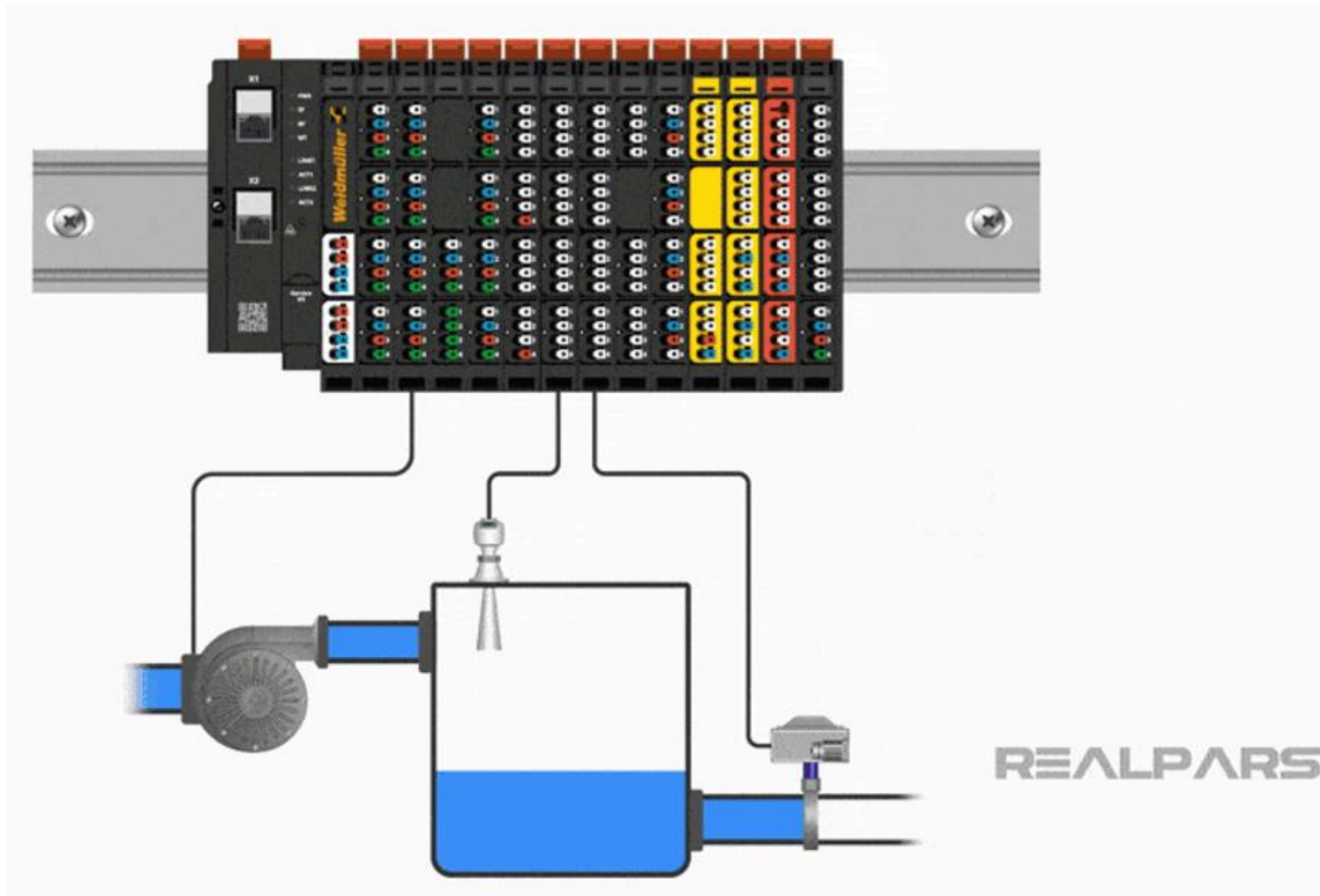
Next Generation Controllers (IIoT Enabled)



Here is an example of an IIoT enabled PLC from Weidmüller. It usually has similar physical components of a typical PLC, which consists of CPU, power modules, IO modules (Digital/ Analog), Communication modules, and safety modules.

In addition to a typical PLC functionality, this IIoT enabled PLC has additional capability to enable OT data to be integrated with IT/ Cloud services for monitoring, as well as, descriptive, preventive, and predictive analytics

Next Generation Controllers (IIoT Enabled)



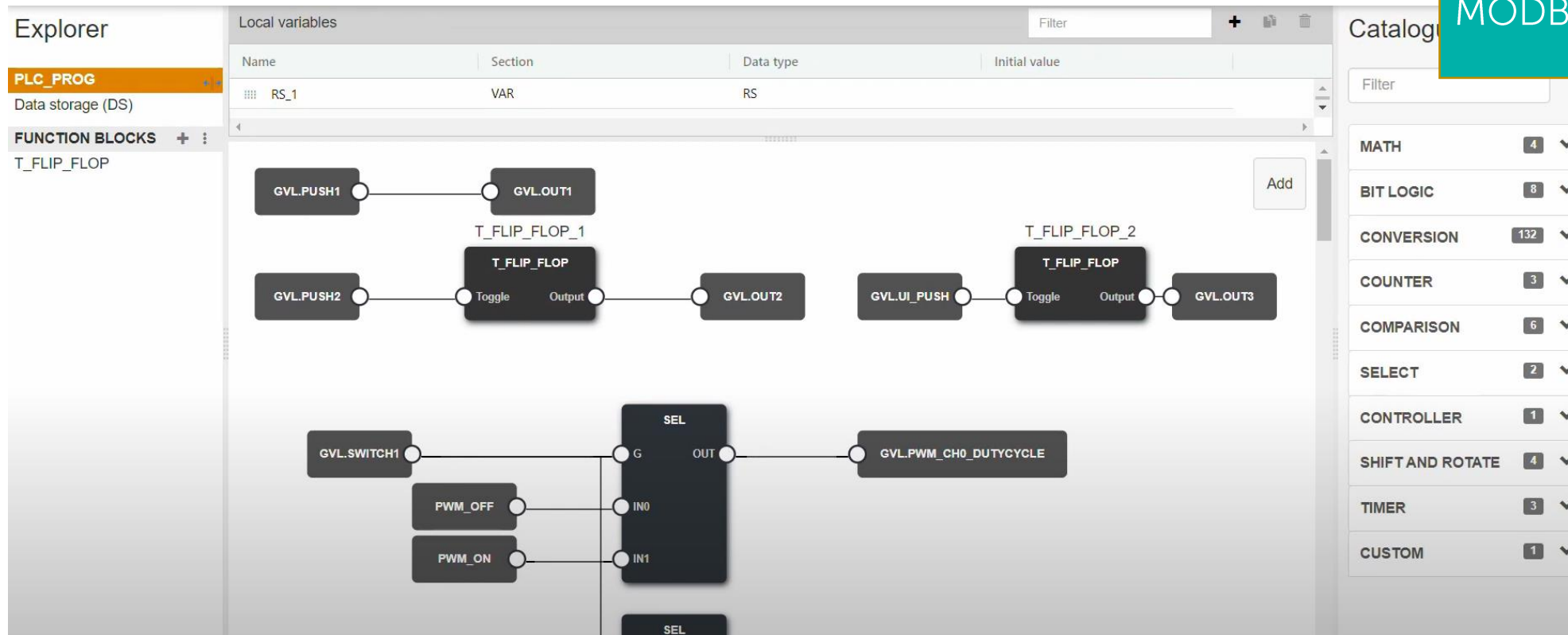
IIoT enabled PLC, provides the flexibility of utilizing the same trusted transmitters or actuators, that are connected using familiar Digital/Analog IO modules.

Examples of common Transmitters: Emerson Rosemount, Yokogawa EJA, Honeywell ST3000, ABB 2600T, etc...

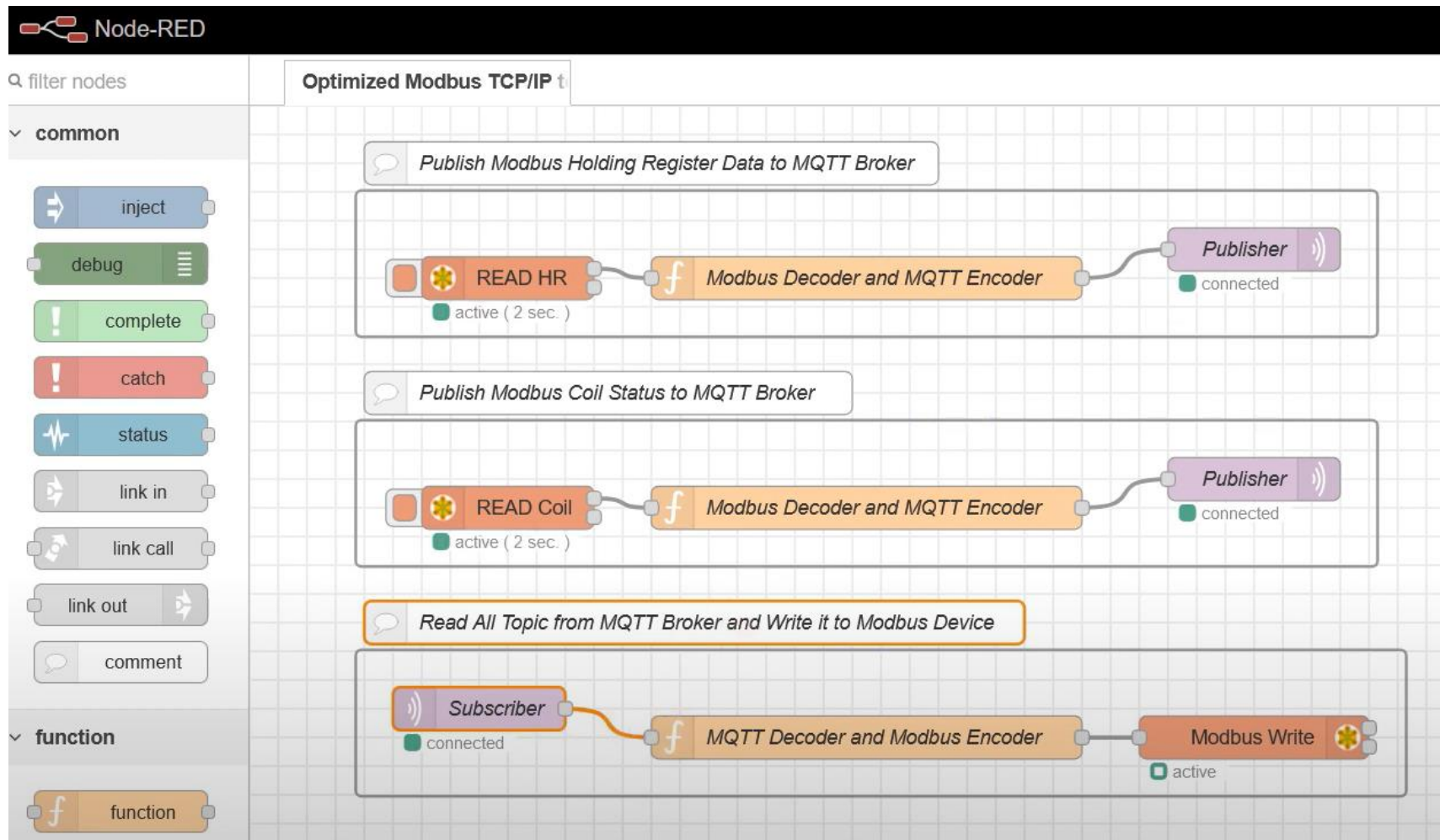
Examples of common Actuators: Emerson Bettis, Honeywell Control Valves, ABB Actuators, Yokogawa's AVP valve positioners, etc...

Next Generation Controllers (IIoT Enabled)

It can be programmed using common ladder diagram and/or functional blocks, where as an example, MODBUS RTU/TCP is used.



Next Generation Controllers (IIoT Enabled)



At the same time, it has the capability to connect to Next Generation Sensors, using built-in middleware IIoT (such as Node-Red) application utilizing IIoT protocols (Ex. OPC-UA, MQTT).

While simultaneously enabling the capability of taking telemetry data from modbus, processing it, and sending the collected data into a new IIoT communication protocol like MQTT or OPC-UA for further monitoring and/or analytics functionality.

03. Key Technologies of IIoT

- Next Generation Sensors
- IIoT Gateways
- IIoT Edge Compute Devices
- Next Gen Controllers (IIoT Enable)
- **Cloud Services**
- IIoT Connectivity (IIoT Protocol)
- IIoT Integration Middleware

Cloud Services

Current OT Implementation

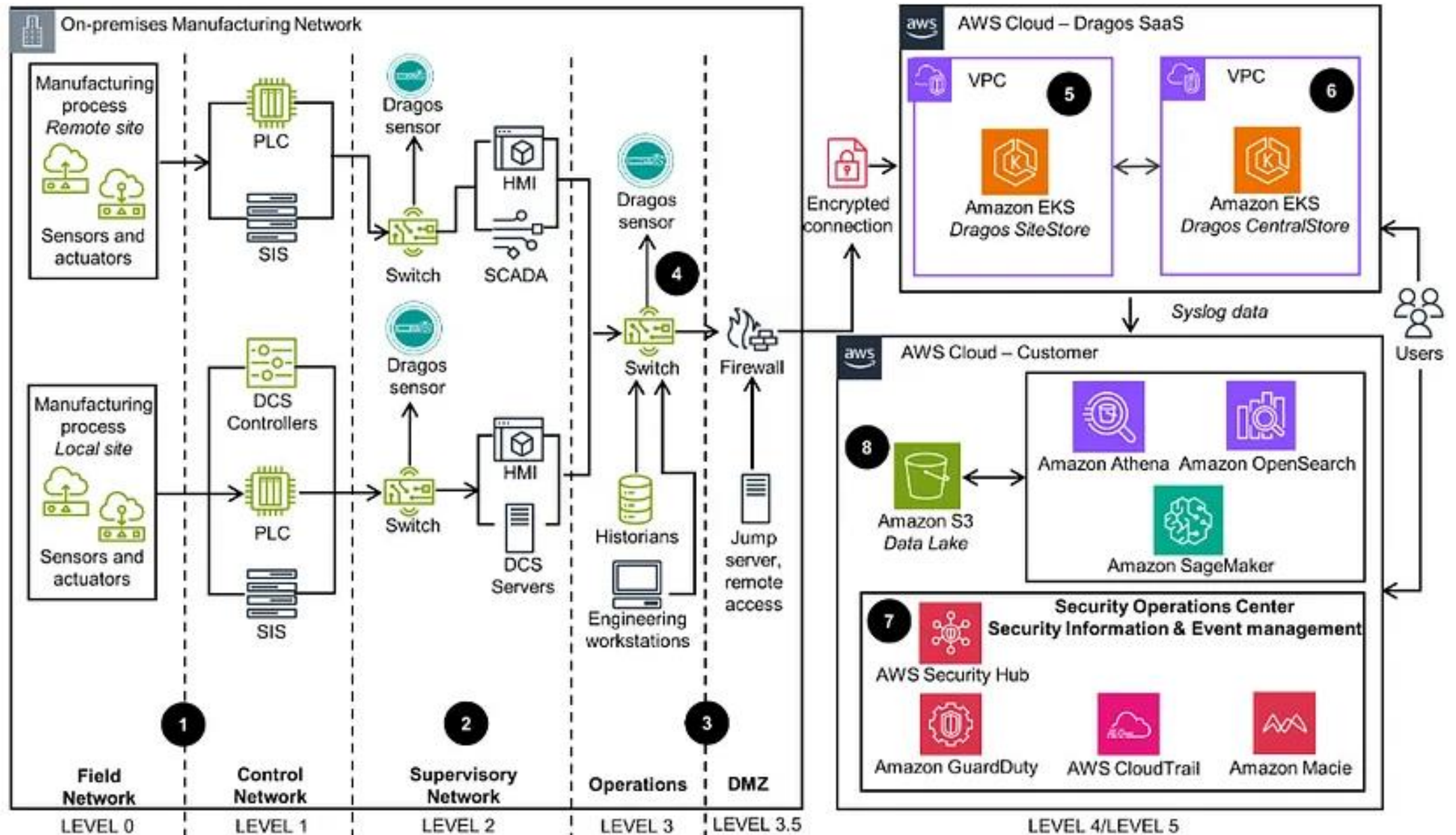
Focus:

- Rely on local SCADA systems, historians, and proprietary networks for data storage and monitoring.
- Designed for localized control and monitoring with limited remote access.

Capabilities:

- Data is typically stored and processed on-site, with limited integration across geographically dispersed assets.
- Heavy reliance on manual data extraction and periodic reporting.

Cloud Services



Cloud Services

Common IIoT Cloud Services

Focus:

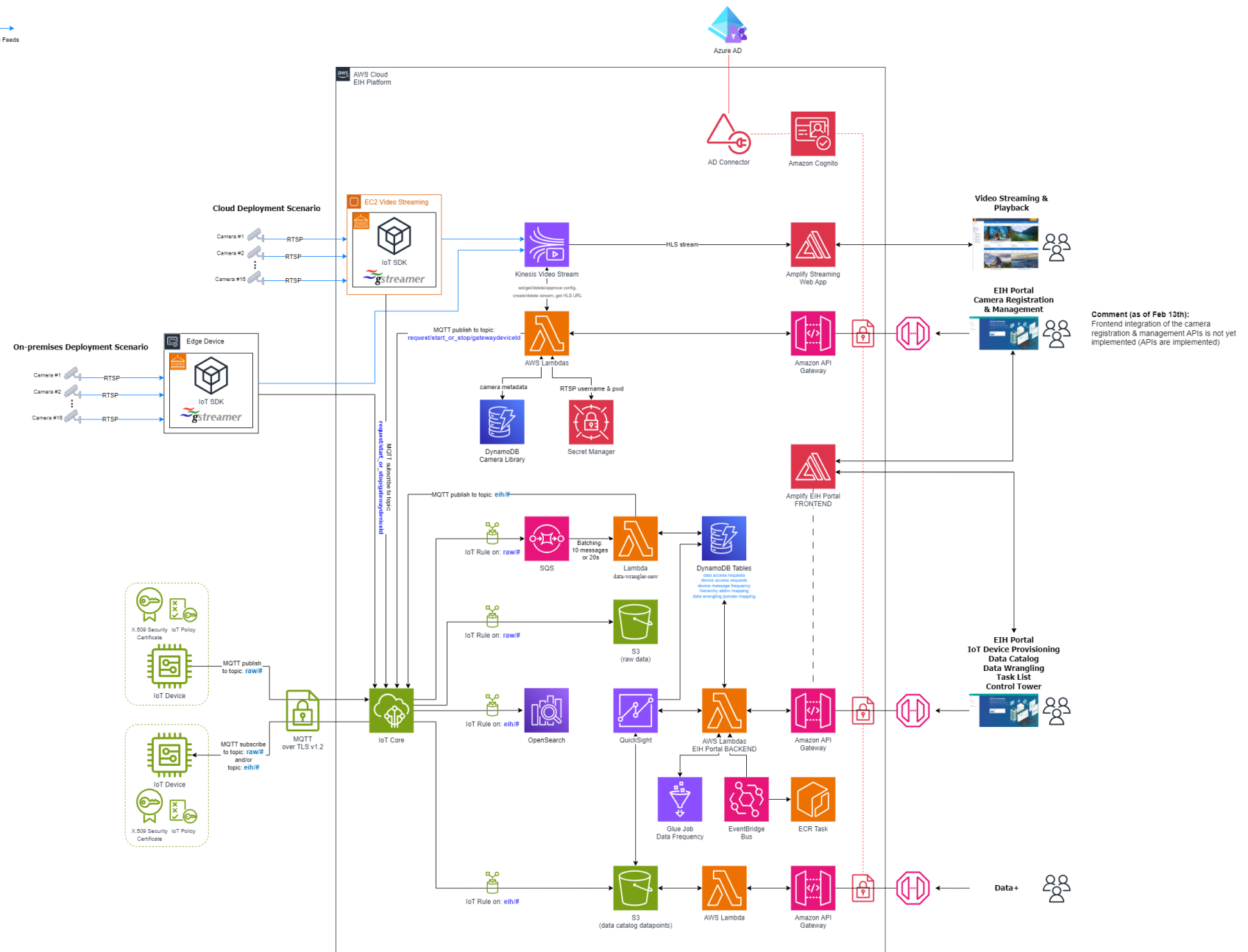
- Provide remote, centralized data aggregation and advanced analytics across multiple locations.
- Enable real-time monitoring, machine learning, and predictive maintenance through scalable, cloud-hosted platforms.

Capabilities:

- Seamless integration of data from edge devices and controllers across sites.
- Scalable processing power and storage to handle large datasets.
- Advanced security features to protect data in transit and at rest.
- Mobile and web-based dashboards for on-demand insights and decision-making.

Cloud Services

Video Feeds



Cloud Services

Examples:

- AWS IoT Core, Microsoft Azure IoT, Software AG, PTC ThingWorx, Siemens, ABB

Key Advantage:

- Enhanced Data-Driven Decision-Making:- Real-time, centralized insights empower proactive maintenance and operational optimization.
- Remote Accessibility & Scalability:- Cloud-hosted platforms offer anywhere access to centralized analytics and dashboards, ensuring faster response times and streamlined operations.
- Improved Integration:- Bridges the gap between field operations and enterprise IT, enabling seamless multi-site connectivity and smarter, connected workflows.

03. Key Technologies of IIoT

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- IIoT Integration Middleware

IIoT Connectivity

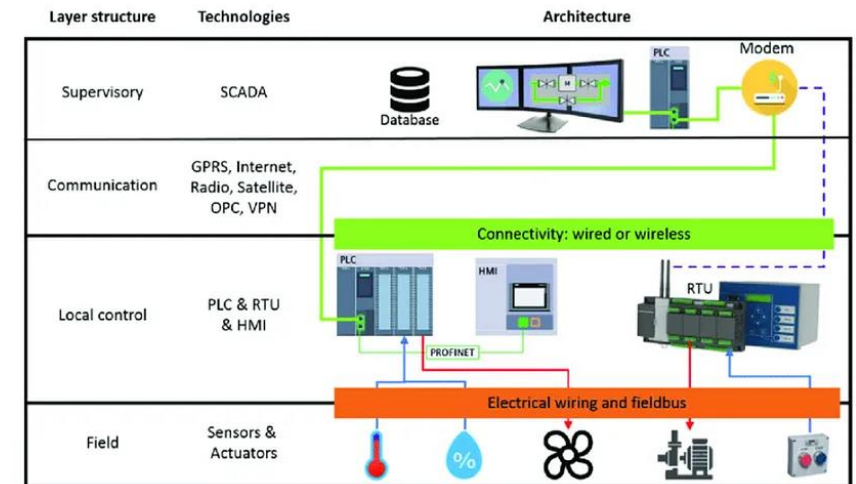
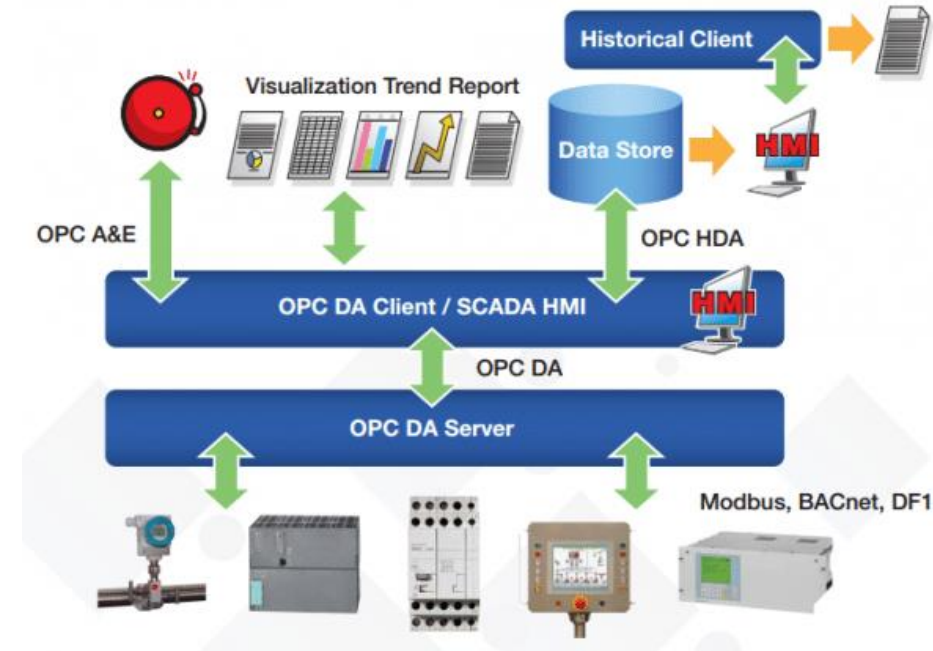
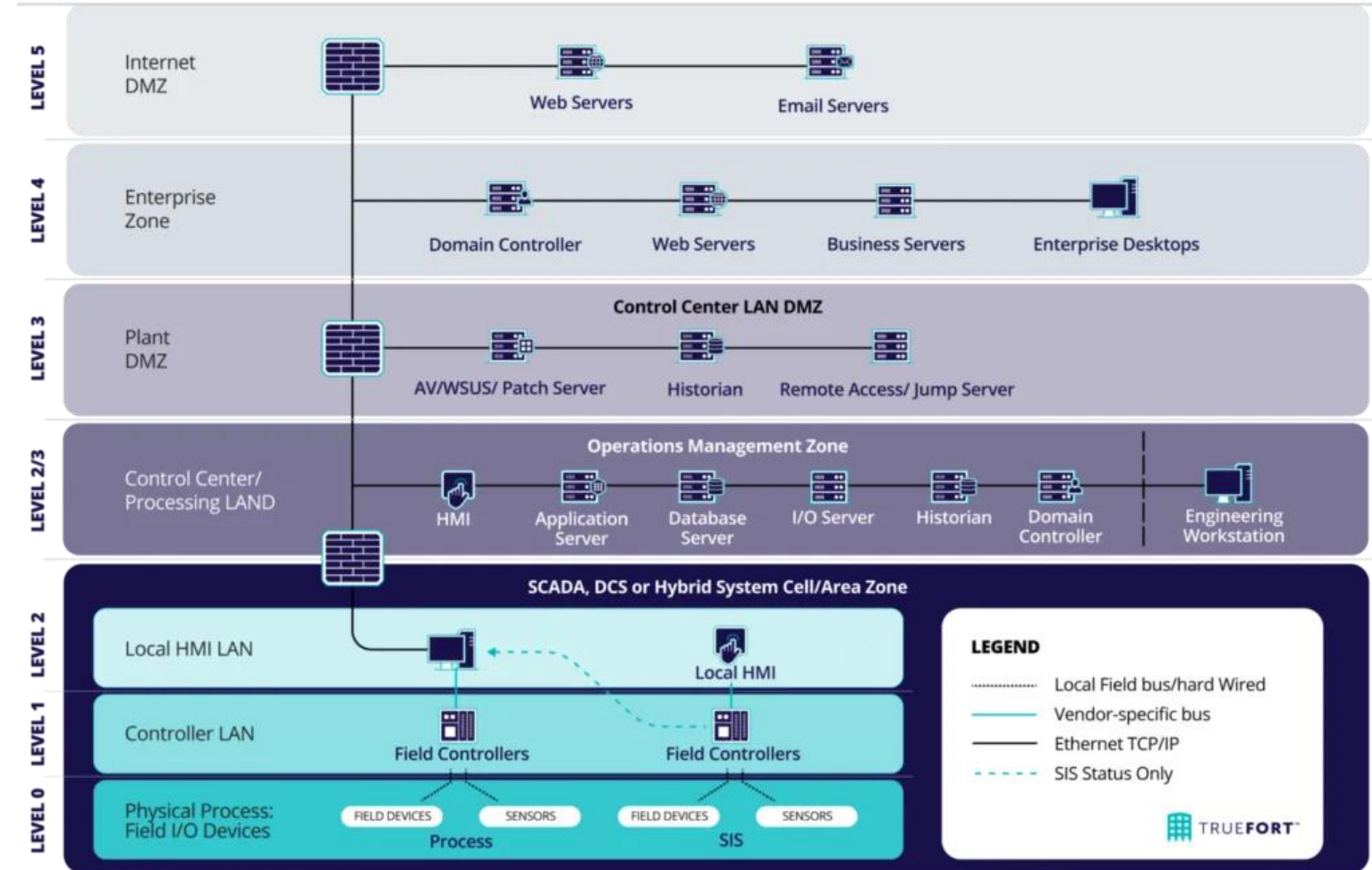
Common OT Protocols

- Purpose & Role:
 - Provide reliable, time-tested communication in oil & gas operations.
 - Primarily used for real-time control and monitoring.
- Common Examples:
 - Modbus (RTU/TCP): Widely deployed in legacy systems for simple device communication.
 - OPC DA (Data Access) part of OPC Classic: Used for real-time data exchange between field devices and SCADA systems.
 - DNP3: Employed in certain SCADA systems, especially in utilities.
- Challenges:
 - Limited security features (e.g., no authentication/encryption)
 - Difficulty integrating with modern IT systems and cloud services.

IIoT Connectivity

Common OT Protocols

The Purdue Model for Industrial Control System (ICS) Security



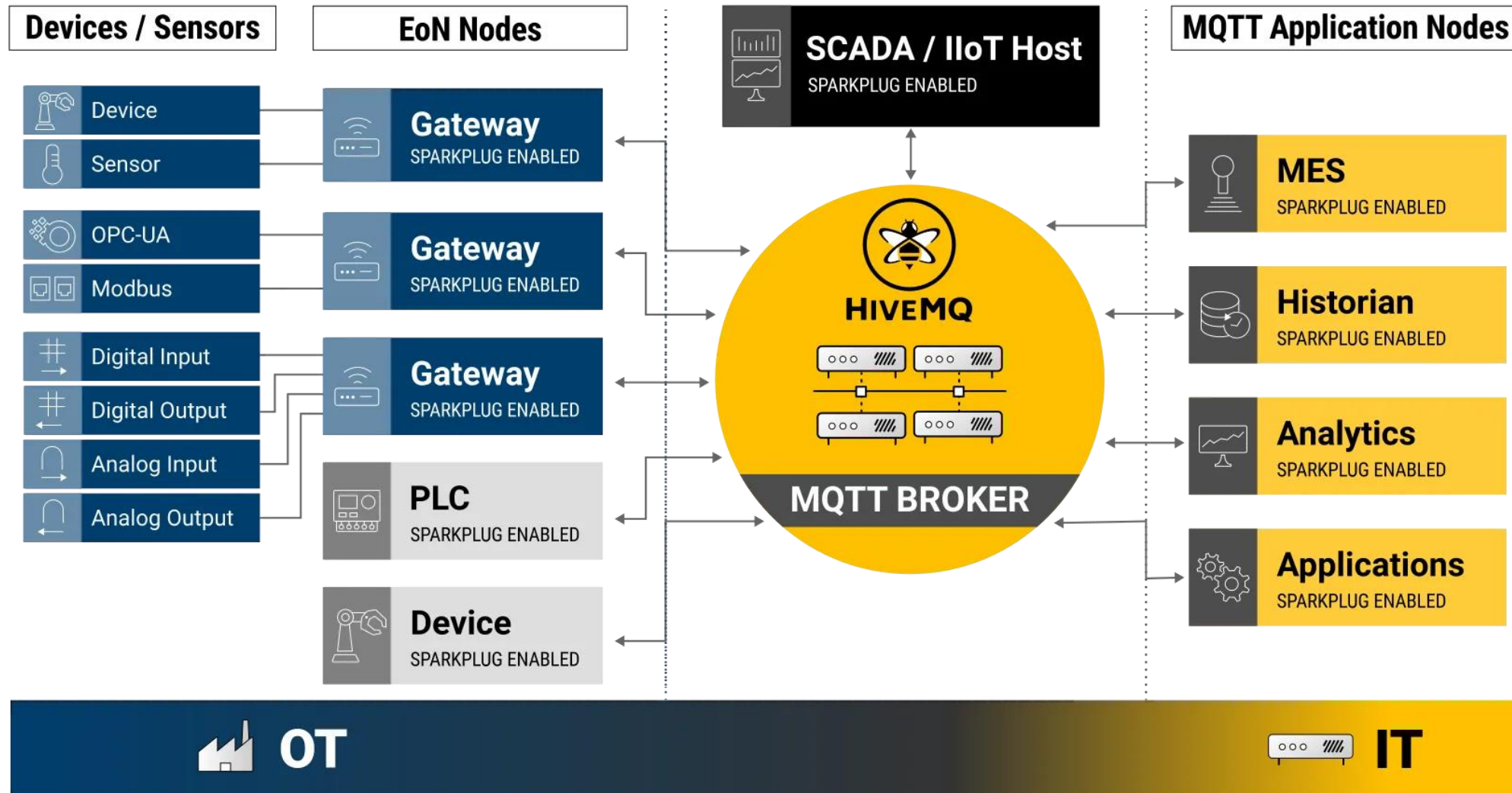
IIoT Connectivity

IIoT Protocols

- Purpose & Role:
 - Enable secure, flexible, and scalable data exchange across OT, IIoT, and IT environments. Support advanced analytics, remote monitoring, and cloud integration.
- Common Examples:
 - OPC UA: Provides secure, cross-platform communication and advanced data modeling.
 - MQTT: A lightweight, publish/subscribe messaging protocol ideal for real-time, remote monitoring.
 - AMQP / CoAP: Facilitate robust and flexible data messaging for enterprise and resource-constrained environments.
 - RTMP / RTSP: Enable Live Video Streaming
- Key Advantages:
 - Enhanced security with built-in encryption and authentication.
 - Improved interoperability between legacy and modern systems.
 - Real-time data processing and integration with cloud-based analytics.

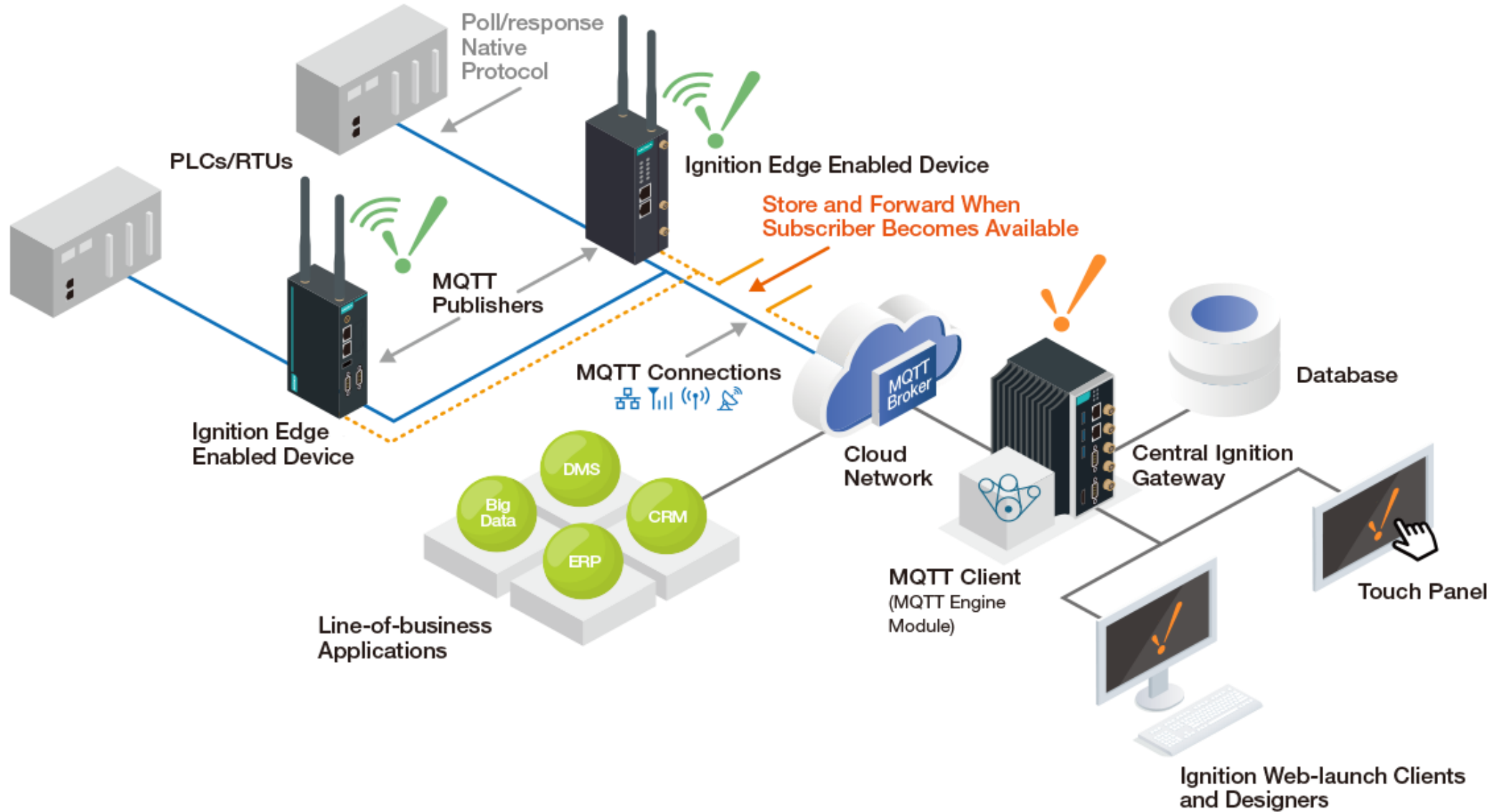
IIoT Connectivity

IIoT Protocols



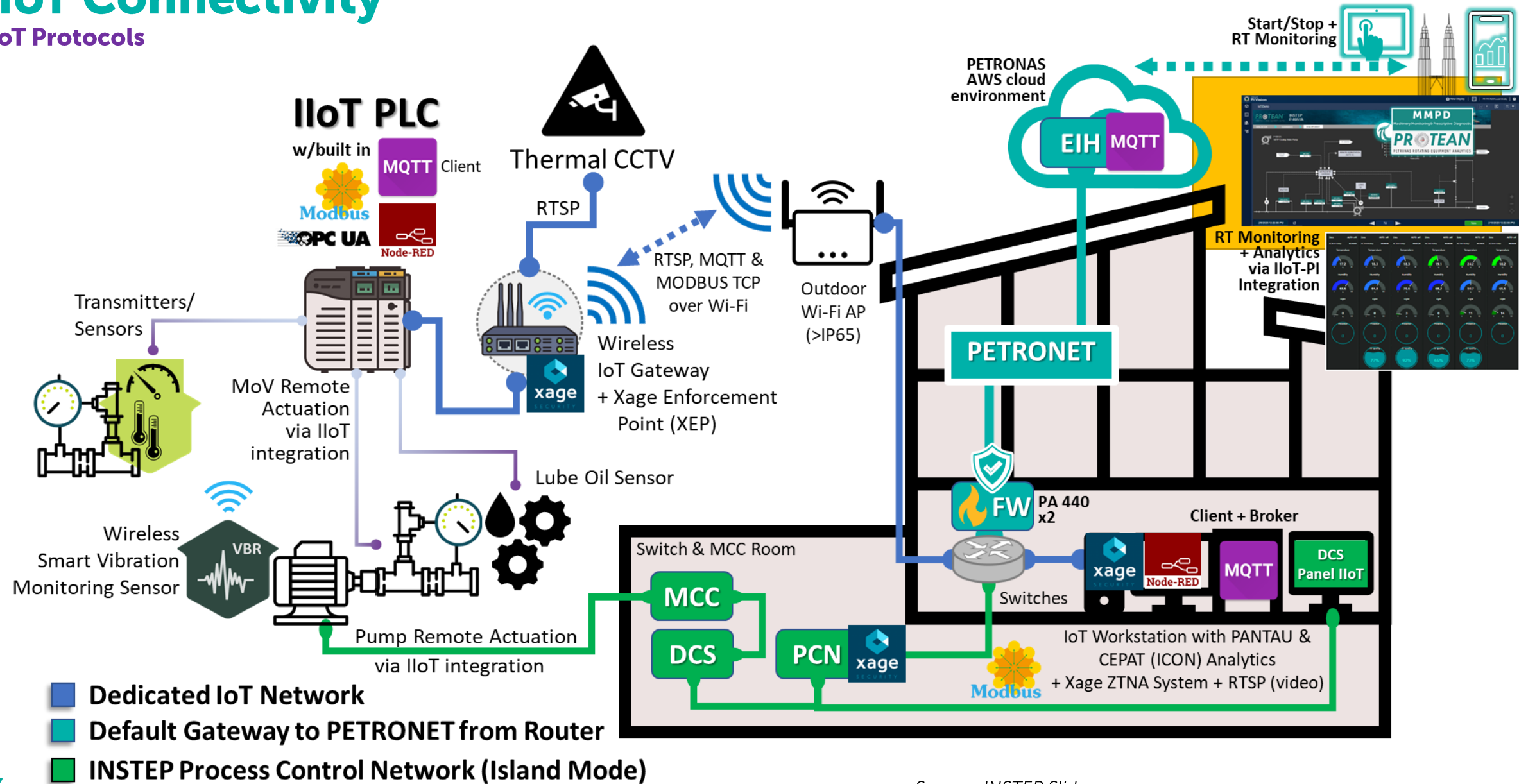
IIoT Connectivity

IIoT Protocols



IIoT Connectivity

IIoT Protocols



Source: INSTEP Slide

IIoT Connectivity

IIoT Protocols

- Bridging the Gap:
- Integration Strategies:
 - Use of modern IIoT gateways and protocol converters allows seamless data flow between traditional OT protocols and IIoT systems.
 - This approach protects existing investments in legacy equipment while unlocking the benefits of advanced connectivity and analytics.
- Outcome for Operations:
 - Safer, more efficient field operations with better decision-making support.
 - Enhanced productivity through real-time data access and predictive maintenance capabilities.

03. Key Technologies of IIoT

- Next Generation Sensors
- IIoT Gateways
- IIoT Edge Compute Devices
- Next Gen Controllers (IIoT Enable)
- Cloud Services
- IIoT Connectivity (IIoT Protocol)
- **IIoT Integration Middleware**

IIoT Integration Middleware (OT)

Focus:

- Use of point-to-point connectivity and proprietary interfaces for data exchange.
- Often reliant on custom programming and manual integration efforts between legacy systems.

Limitations:

- Scalability challenges and increased complexity when integrating diverse systems.
- Limited real-time data visibility and interoperability across platforms.

IIoT Integration Middleware

Focus:

- Serve as a bridging layer that seamlessly connects legacy OT systems with modern IIoT and cloud applications.
- Facilitate multi-protocol integration, enabling secure, real-time data exchange across disparate environments.

Capabilities:

- **Protocol Translation:** Converts data from legacy OT protocols (e.g., Modbus, OPC DA) into formats suitable for IIoT applications (e.g., MQTT, OPC UA).
- **Data Aggregation & Processing:** Provides real-time data collection, filtering, and transformation for actionable insights.
- **Interoperability:** Ensures smooth integration between OT devices and enterprise-level systems, reducing the need for extensive custom coding.

IIoT Integration Middleware

Examples:

- **Kepware:** An industrial connectivity platform that provides standardized communication across diverse devices.
- **Ignition:** A unified platform that supports SCADA, IIoT, MES, and real-time analytics.
- **Node-RED:** An open-source flow-based tool for wiring together hardware devices, APIs, and online services.
- **ThingWorx:** An IIoT platform that accelerates application development and real-time integration of connected devices.

IIoT Integration Middleware

Key Advantage:

- **Enhanced Interoperability & Flexibility:**
 - Streamlines integration between legacy OT systems and modern IIoT/cloud platforms.
 - Reduces complexity and accelerates digital transformation initiatives.
- **Real-Time Data Access:**
 - Enables rapid decision-making through centralized, actionable insights and advanced analytics.
- **Scalability & Security:**
 - Provides a secure, scalable framework for multi-protocol data exchange that supports current and future operational needs.

